Putnam - Solutions

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Contents

Chapter 1

Putnam 1938

A1

Let $A(k) = ak^3 + bk^2 + ck + d$, where $d = M$ since $A(0) = M$. Let us compute the volume by integrating $A(k)$:

$$V = \int_{-\frac{h}{2}}^{\frac{h}{2}} A(k)dk$$

$$= \left[\frac{a}{4}k^4 + \frac{b}{3}k^3 + \frac{c}{2}k^2 + Mk\right]_{-\frac{h}{2}}^{\frac{h}{2}}$$

$$= \frac{bh^3}{12} + Mh$$

Our goal is to get

$$V = \frac{h(B + 4M + T)}{6}$$

Notice that

$$B = A\left(-\frac{h}{2}\right) = -a\frac{h^3}{8} + b\frac{h^2}{4} - c\frac{h}{2} + M$$

$$T = A\left(\frac{h}{2}\right) = a\frac{h^3}{8} + b\frac{h^2}{4} + c\frac{h}{2} + M$$

thus

$$B + T = \frac{bh^2}{2} + 2M$$

Substituting into our goal, we simplify

$$V = \frac{h(B + 4M + T)}{6} = \frac{h(\frac{bh^2}{2} + 6M)}{6} = \frac{bh^3}{12} + Mh$$

Hence, the volume formula is indeed correct.

We can use this formula to derive the formula for a cone and sphere. For a cone with base radius r and height h, the area formula is

$$A(k) = \pi \left(\frac{r}{h} \left(k + \frac{h}{2} \right) \right)^2$$

thus a 2nd degree polynomial, which satisfies the requirements for using the special formula, hence

$$\begin{aligned} V &= \frac{h(B + 4M + T)}{6} \\ &= \frac{h(\pi r^2 + 4\pi \left(\frac{r}{2}\right)^2 + 0)}{6} \\ &= \frac{h(2\pi r^2)}{6} = \frac{\pi r^2 h}{3} \end{aligned}$$

For a sphere with radius $r = \frac{h}{2}$, the cross-sectional area is

$$A(k) = \pi \left(\frac{h^2}{4} - k^2 \right)$$

thus again a 2nd degree polynomial in k, so we apply the special formula to get

$$\begin{aligned} V &= \frac{h(B + 4M + T)}{6} \\ &= \frac{h(0 + 4\pi \frac{h^2}{4} + 0)}{6} = \frac{\pi h^3}{6} \end{aligned}$$

Expressed in radius $2r = h$, we get

$$V = \frac{4}{3}\pi r^3$$

A2

$$V = 2 \cdot \frac{\pi r^2 h}{3} + \pi r^2 h = \frac{5}{3}\pi r^2 h$$
$$A = 2\pi r h + 2\pi r \sqrt{r^2 + h^2}$$

We want to express either r or h in terms of A and the other:

$$\begin{aligned} A - 2\pi r h &= 2\pi r \sqrt{r^2 + h^2} \\ A^2 - 4A\pi r h + 4\pi^2 r^2 h^2 &= 4\pi^2 r^2 (r^2 + h^2) \\ A^2 - 4A\pi r h + 4\pi^2 r^2 h^2 &= 4\pi^2 r^4 + 4\pi^2 r^2 h^2 \\ A^2 - 4A\pi r h &= 4\pi^2 r^4 \end{aligned}$$

$$h = \frac{A^2 - 4\pi^2 r^4}{4A\pi r}$$

Substituting into the volume equation, we get

$$V = \frac{5}{3}\pi r^2 \frac{A^2 - 4\pi^2 r^4}{4A\pi r}$$

$$= \frac{5}{12}\frac{r(A^2 - 4\pi^2 r^4)}{A}$$

We want to maximize this equation given a fixed A, so we can ignore the coefficient and denominator, leaving only

$$r(A^2 - 4\pi^2 r^4) = A^2 r - 4\pi^2 r^5$$

Setting the derivative to zero, we get

$$A^2 - 20\pi^2 r^4 = 0$$

$$\Rightarrow r = \sqrt{\frac{A}{\sqrt{20}\pi}}$$

Note that, from the height equation, we know that

$$0 < A^2 - 4\pi^2 r^4$$

$$\Rightarrow r < \sqrt{\frac{A}{2\pi}}$$

$\sqrt{20} > \sqrt{16} = 4 > 2$, so the r we found is safely within the bounds.

A3

Velocities are

$$\frac{dx}{dt} = 3t^2 - 1$$

$$\frac{dy}{dt} = 4t^3 + 1$$

I think they are asking for the speed, which is

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = \sqrt{16t^6 + 9t^4 + 8t^3 - 6t^2 + 2}$$

Since we are finding the maximum, we can ignore the square root. Since a polynomial is differentiable everywhere, its maxima happen when derivative is zero

$$96t^5 + 36t^3 + 24t^2 - 12t = 0$$

Clearly, $t = 0$ is an extrema (or saddle). To know whether it's a maximum, we can take the second derivative at 0

$$480t^4 + 108t^2 + 48t - 12$$

which equals -12 (negative) when t are replaced with 0, hence $t = 0$ is a local maximum.

To know whether it's an inflection point, we can compute $\frac{d^2y}{dx^2}$ at $t = 0$, which should be possible since the tangent is not vertical (specifically, $\vec{v}(0) = (-1, 1)$):

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3t^2 - 1}{4t^3 + 1}$$

$$\frac{d}{dx}\frac{dy}{dx} = \frac{d}{dt}\frac{dy}{dx} \cdot \frac{dt}{dx} = \frac{d}{dt}\frac{dy}{dx} \cdot \frac{1}{\frac{dx}{dt}}$$

$$= \frac{(4t^3 + 1)(6t) - (3t^2 - 1)(12t^2)}{(4t^3 + 1)^2} \cdot \frac{1}{3t^2 - 1}$$

$$= \frac{-12t^4 + 12t^2 + 6t}{(4t^3 + 1)^2(3t^2 - 1)}$$

Substituting 0, the whole thing becomes 0. Hence, the second derivative is zero, but we still need to know whether there is a sign change for the second derivative (which represents change in concavity).

For t near 0, the denominator is always negative. However, the derivative of the numerator is 6 (positive) at $t = 0$, and since the value of the numerator is 0 at $t = 0$, the numerator changes sign when $t < 0$ becomes $t > 0$. Hence, there is a sign change in the second derivative at $t = 0$, hence it's an inflection point.

A4

I don't understand the problem.

A5

(1) Using L'Hopital's,

$$\lim_{x\to\infty} \frac{x^2}{e^x} = \lim_{x\to\infty} \frac{2x}{e^x} = \lim_{x\to\infty} \frac{2}{e^x} = 0$$

Alternatively, we can argue exponential growth dominates polynomial growth.
(b) Since

$$\lim_{x\to 0}(1 + \sin 2x)^{\frac{1}{x}}$$

$$= e^{\lim_{x\to 0} \frac{1}{x} \log(1+\sin 2x)}$$

with L'Hopital's,

$$= e^{\lim_{x\to 0} \frac{2\cos 2x}{1+\sin 2x}} = e^2$$

is finite. Inside $[0, \infty)$, it's the only undefined point of the integrand, which has a finite limit, hence the integral is bounded for every k. Hence, as $k \to 0$, the integral goes to 0. Therefore, we can apply L'Hopital's

$$\lim_{k\to 0} \frac{\int_0^k (1+\sin 2x)^{\frac{1}{x}}\, dx}{k}$$

$$= \lim_{k\to 0} (1+\sin 2k)^{\frac{1}{k}}$$

which we found to be equal to e^2.

A6

Suppose the pool is a 1x1 square. Let s, r be swimming and running speeds respectively. If $s > r$, then the obvious solution is to swim diagonally directly to the opposite corner. If $s < r$, there are two cases.

If swimming the hypothenuse is slower than running both edges, i.e. $\frac{\sqrt{2}}{s} > \frac{2}{r} \Rightarrow s < \frac{r}{\sqrt{2}}$, then the solution is to run along the edges.

If she starts swimming at the start, there are three options. Reaching the bottom or left edges, reaching the top or right edges, or reaching the opposite corner. The latter, by our initial assumption $\frac{\sqrt{2}}{s} > \frac{2}{r}$, is obviously slower. Obviously, there's a fourth option to reach somewhere else in the water, but she will eventually reach one of those three options, and this fourth option checkpoint would simply be a point in her path to one of those three options.

If she lands on the bottom or left edges, then she effectively used a longer path than simply running directly to the landing point. Hence, this option is discarded.

If she lands on the top or right edges, she has many options. Going back to the starting point or to the bottom or left edge is stupidly wrong. The remaining options are landing back onto the top or right edge, or simply reaching the opposite corner. Re-landing onto the top or right edge is, again, a longer path than simply running along the edge. Hence, her only viable option is running directly to the opposite corner.

Depending on her landing ordinate $0 < y < 1$ on the right edge (the reasoning is equivalent for the top edge), her time is

$$T = \frac{\sqrt{1+x^2}}{s} + \frac{1-x}{r} > \frac{\sqrt{1+x^2}}{\frac{r}{\sqrt{2}}} + \frac{1-x}{r}$$

$$= \frac{1-x+\sqrt{2}\sqrt{1+x^2}}{r}$$

The numerator is always greater or equal to 2, because

$$\begin{aligned} \sqrt{2}\sqrt{1+x^2} - x + 1 &\geq 2 \\ \Rightarrow \sqrt{2}\sqrt{1+x^2} &\geq x+1 \\ \Rightarrow 2(1+x^2) &\geq x^2+2x+1 \\ \Rightarrow x^2+1 &\geq 2x \\ \Rightarrow (x-1)^2 &\geq 0 \end{aligned}$$

is true.

Hence, $T > \frac{2}{r}$, thus running along the edges is always faster if swimming the hypothenuse is slower than running the edges, if she starts by swimming.

Another case is when $\frac{r}{\sqrt{2}} \leq s < r$, i.e. swimming the hypothenuse is faster than running the edges, but running speed is still higher than swimming speed. Going back to our three options,

A7

(1) Simple double integration exercise.
(2) Using the symmetry of the surface $z = xy$, we can reduce to three cases for the line, when viewed from above:

- passes through the center
- crosses either x or y axis
- crosses both x and y axes

Starting with the first case, if the line passes through the center, then $y = kx$ for some k, whence $z = xy = x(kx) = kx^2$. This is only a line when $k = 0$, where we end up with the line $y = z = 0$ varying with x. The only case missing is the vertical line case, where $x = 0$. This is obviously a line, since $z = xy = 0$. Hence $x = z = 0$ with only y varying, which is a line.

For the second case, if a line crosses only one axis, then it is parallel with that axis (when viewed from above). Suppose $x = x_0$ is fixed, then $z = x_0 y$, which is a line with slope x_0 and independent variable y. Same thing when y is fixed. Hence, we have an infinite set of such lines. The first case is actually a special case, where $x_0 = 0$ or $y_0 = 0$.

Finally, if a line crosses both the x and y axes, then $x = 0$ and $y = 0$ reach occur once, whence $z = xy = 0$ at those two points, hence the line must lay flat against the horizontal plane, i.e. $z = 0$ everywhere. However, everywhere else the line passes through, the plane assumes value $z \neq 0$, hence it cannot be a line.

To summarize, we have the lines:

- $z = x_0 y$, $x = x_0$ for $x_0 \in \mathbb{R}$
- $z = y_0 x$, $y = y_0$ for $y_0 \in \mathbb{R}$

B1

(1) Solution found online. Consider the cofactor matrix A_{ij} and the transpose of the original matrix A^T.

$$\begin{bmatrix} A_{11} & A_{12} & A_{13} & A_{14} \\ A_{21} & A_{22} & A_{23} & A_{24} \\ A_{31} & A_{32} & A_{33} & A_{34} \\ A_{41} & A_{42} & A_{43} & A_{44} \end{bmatrix} \begin{bmatrix} a_{11} & a_{21} & a_{31} & a_{41} \\ a_{12} & a_{22} & a_{32} & a_{42} \\ a_{13} & a_{23} & a_{33} & a_{43} \\ a_{14} & a_{24} & a_{34} & a_{44} \end{bmatrix}$$

If we compute the matrix product, we notice that

$$a_{i1}A_{1i} + a_{i2}A_{2i} + a_{i3}A_{3i} + a_{i4}A_{4i} = \det A = d$$

but

$$a_{i1}A_{1j} + a_{i2}A_{2j} + a_{i3}A_{3j} + a_{i4}A_{4j} = 0$$

for $i \neq j$, because it's equivalent to the determinant of A where the j-th row is replaced with the i-th row. Hence, the resulting matrix is

$$\begin{bmatrix} d & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & 0 & d & 0 \\ 0 & 0 & 0 & d \end{bmatrix}$$

which has determinant d^4. Since it was the result of multiplying the transpose with the cofactor matrix, and the transpose has determinant d, then the cofactor matrix itself has determinant $\frac{d^4}{d} = d^3$.

(2) If $P(x_1) = x_1$, then $P(P(x_1)) = P(x_1) = x_1$, hence $P(P(x_1)) = x_1$, so x_1 is a root of $P(P(x)) = x$. Same for the other root x_2.

I'm too dumb for the rest.

B2

Solution found online.

$$\begin{aligned} yy'' &= 2y'y' \\ \Rightarrow \frac{y''}{y'} &= 2\frac{y'}{y} \\ \Rightarrow \log y' &= 2\log y + C \\ \Rightarrow y' &= Cy^2 \\ \Rightarrow C\frac{dy}{y^2} &= dx \\ \Rightarrow -\frac{C}{y} &= x + D \end{aligned}$$

Since it passes through $x = 1, y = 1$, then $-C = 1 + D$. Hence, we have the family of solutions

$$\frac{1+D}{x+D} = y$$

B3

Using the disk's reference frame, the insect is simply turning at a constant speed and advancing at a constant speed. Hence, its path is an arc of a bigger circle. The disk's edge and the big circle intersect at right angles, hence the insect's path touches the disk's edge at right angle. Since at the end of its path, the insect is facing north, the insect touches the edge again at the disk's northernmost point.

B4

Using algebra, we can find the parabola's equation:

$$x^2 + \left(\frac{m}{2} - y\right)^2 = \left(\frac{m}{2} + y\right)^2$$

$$x^2 + y^2 - my + \frac{m^2}{4} = y^2 + my + \frac{m^2}{4}$$

$$y = \frac{x^2}{2m}$$

assuming the parabola passes through the origin. The derivative is thus

$$y' = \frac{x}{m}$$

This is the slope of the tangent line. For the perpendicular line, the slope is $-\frac{m}{x}$. By symmetry, we can restrict ourselves to using $x < 0$, hence the lines have positive slope $\frac{m}{x_0}$ where x_0 is the horizontal distance from the origin. Using algebra, we can find the line's equation:

$$b = \frac{m}{x_0} \cdot x_0 + \frac{x_0^2}{2m} = m + \frac{x_0^2}{2m}$$

$$\Rightarrow y = \frac{m}{x_0}x + m + \frac{x_0^2}{2m}$$

Equating with the parabola, we get

$$\frac{m}{x_0}x + m + \frac{x_0^2}{2m} = \frac{x^2}{2m}$$

$$0 = \frac{x^2}{2m} - \frac{m}{x_0}x - m - \frac{x_0^2}{2m}$$

$$= x_0x^2 - 2m^2x - 2m^2x_0 - x_0^3$$

The quadratic formula gives

$$x = \frac{m^2}{x_0} + \sqrt{\frac{m^4}{x_0^2} + 2m^2 + x_0^2}$$

which can be simplified

$$x = \frac{m^2}{x_0} + \sqrt{\left(\frac{m^2}{x_0} + x_0\right)^2} = \frac{2m^2}{x_0} + x_0$$

The corresponding y value is

$$\frac{2m^3}{x_0^2} + 2m + \frac{x_0^2}{2m}$$

The distance is

$$\sqrt{\left(\frac{2m^3}{x_0^2} + 2m + \frac{x_0^2}{2m} - \frac{x_0^2}{2m}\right)^2 + \left(x_0 + \frac{2m^2}{x_0} + x_0\right)^2}$$

$$= \sqrt{\left(\frac{2m^3}{x_0^2} + 2m\right)^2 + \left(2x_0 + \frac{2m^2}{x_0}\right)^2}$$

$$= \sqrt{\frac{4m^6}{x_0^4} + \frac{12m^4}{x_0^2} + 12m^2 + 4x_0^2}$$

$$= \frac{2}{x_0^2}\sqrt{m^6 + 3m^4x_0^2 + 3m^2x_0^4 + x_0^6}$$

$$= \frac{2}{x_0^2}(m^2 + x_0^2)^{\frac{3}{2}}$$

We can thus minimize it with respect to some $z = x_0^2$:

$$\frac{(m^2 + z)^{\frac{3}{2}}}{z}$$

The derivative is

$$\frac{\frac{3}{2}z(m^2 + z)^{\frac{1}{2}} - (m^2 + z)^{\frac{3}{2}}}{z^2}$$

Setting it to zero, we get

$$\frac{3}{2}z(m^2 + z)^{\frac{1}{2}} = (m^2 + z)^{\frac{3}{2}}$$

$$\frac{3}{2}z = m^2 + z$$

$$z = 2m^2$$

Plugging it back into the distance, we get a minimum distance of

$$\frac{2}{2m^2}(m^2 + 2m^2)^{\frac{3}{2}} = \frac{(3m^2)^{\frac{3}{2}}}{m^2} = \sqrt{27}m = 3\sqrt{3}m$$

B5

The hyperbola's equation has the form

$$x^2 - y^2 = c$$

assuming it's horizontal. The vertical hyperbola would simply exchange x and y, but the solution is similar.

The derivative is

$$\frac{dy}{dx} = \frac{x}{y}$$

The tangent's y-intercept can be found for a given x_0

$$b = y_0 - \frac{x_0^2}{y_0}$$

Let D be the distance from the line to the origin. We can solve for D using the fact that the right triangle formed by the tangent, y-axis, and distance, has the same legs proportion as the tangent's slope:

$$D^2 + \left(\frac{x_0}{y_0} D\right)^2 = b^2$$

$$D^2 = \frac{\left(y_0 - \frac{x_0^2}{y_0}\right)^2}{1 + \frac{x_0^2}{y_0^2}}$$

$$= \frac{\left(y_0^2 - x_0^2\right)^2}{y_0^2 + x_0^2}$$

Let h be the segment representing the distance from the tangent to the origin. For a given counterclockwise angle θ for h from the positive x-axis, the tangent's slope is

$$-\frac{\cos\theta}{\sin\theta}$$

which must correspond to the derivative, hence

$$\frac{x_0}{y_0} = -\frac{\cos\theta}{\sin\theta}$$

$$x_0^2 \sin^2\theta - y_0^2 \cos^2\theta = 0$$

We can derive the appropriate polar formulas for x_0^2 and y_0^2:

$$x_0^2 \sin^2\theta - (x_0^2 - c)\cos^2\theta = 0$$

$$x_0^2 = -\frac{c\cos^2\theta}{\sin^2\theta - \cos^2\theta}$$

and

$$(y_0^2 + c)\sin^2\theta - y_0^2\cos^2\theta = 0$$

$$y_0^2 = -\frac{c\sin^2\theta}{\sin^2\theta - \cos^2\theta}$$

We can rewrite the distance formula using these polar coordinates

$$\begin{aligned} D^2 &= \frac{\left(-\frac{c\sin^2\theta}{\sin^2\theta-\cos^2\theta} + \frac{c\cos^2\theta}{\sin^2\theta-\cos^2\theta}\right)^2}{-\frac{c\sin^2\theta}{\sin^2\theta-\cos^2\theta} - \frac{c\cos^2\theta}{\sin^2\theta-\cos^2\theta}} \\ &= \frac{\left(\frac{c\cos 2\theta}{-\cos 2\theta}\right)^2}{-\frac{c}{-\cos 2\theta}} \\ &= \frac{c^2}{\frac{c}{\cos 2\theta}} \\ &= c\cos 2\theta \end{aligned}$$

Hence, the equation of the locus is

$$r = \sqrt{c\cos 2\theta}$$

which remains undefined for $\theta \in [\frac{\pi}{4}, \frac{3\pi}{4}] \cup [\frac{5\pi}{4}, \frac{7\pi}{4}]$, to match the fact that the tangent is impossible for those angles by graphical inspection. Basically, it's a figure 8 without the center.

Chapter 2

Putnam 1939

A1

Graph is symmetric above and below the x-axis, but the positive gradient is only found on the top. Hence, we have

$$y = \sqrt{x^3} = x^{\frac{3}{2}}$$

The derivative is

$$y' = \frac{3}{2}\sqrt{x}$$

which must be equal to 1, hence

$$\frac{3}{2}\sqrt{x} = 1$$

$$\Rightarrow x = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$$

The path integral formula is

$$\int_0^{\frac{4}{9}} \sqrt{1 + (y')^2} dx$$

which we substitute and solve

$$\int_0^{\frac{4}{9}} \sqrt{1 + \left(\frac{3}{2}\sqrt{x}\right)^2} dx$$

$$= \int_0^{\frac{4}{9}} \sqrt{1 + \frac{9}{4}x}\ dx$$

$$= \left[\frac{4}{9} \cdot \frac{2}{3}\left(1 + \frac{9}{4}x\right)^{\frac{3}{2}}\right]_0^{\frac{4}{9}}$$

$$= \frac{8}{27}\left(1 + \frac{9}{4} \cdot \frac{4}{9}\right)^{\frac{3}{2}} - \frac{8}{27}$$
$$= \frac{8}{27}\left(2\sqrt{2} - 1\right)$$

A2

Derivative is
$$y' = 3x^2$$
The tangent's equation can be found
$$b = y_0 - x_0 \cdot y_0' = x_0^3 - x_0 \cdot 3x_0^2 = -2x_0^3$$
$$y = 3x_0^2 \cdot x - 2x_0^3$$
The intersections with the cubic can be found
$$3x_0^2 \cdot x - 2x_0^3 = x^3$$
$$0 = x^3 - (3x_0^2)x + 2x_0^3$$
$$= (x - x_0)(x - x_0)(x + 2x_0)$$
We already know it intersects at $x = x_0$, hence the desired point is $x = -2x_0$. The derivative at that point equals
$$y' = 3(-2x_0)^2 = 4(3x_0^2)$$
which is exactly four times the derivative $3x_0^2$ at $x = x_0$.

A3

We can express the cubic as a product using its roots
$$y = (x - \alpha)(x - \beta)(x - \gamma)$$
Hence, we can find formulas for a, b, c in terms of these roots
$$= (x^2 - (\alpha + \beta)x + \alpha\beta)(x - \gamma)$$
$$= x^3 + (-\alpha - \beta - \gamma)x^2 + (\alpha\beta + \alpha\gamma + \beta\gamma)x - \alpha\beta\gamma$$
$$\Rightarrow a = -\alpha - \beta - \gamma$$
$$\Rightarrow b = \alpha\beta + \alpha\gamma + \beta\gamma$$
$$\Rightarrow c = -\alpha\beta\gamma$$
Hence, for roots $\alpha^3, \beta^3, \gamma^3$, we have these corresponding a', b', c' values
$$a' = -\alpha^3 - \beta^3 - \gamma^3$$

$$b' = \alpha^3\beta^3 + \alpha^3\gamma^3 + \beta^3\gamma^3$$
$$c' = -\alpha^3\beta^3\gamma^3$$

We want to express each of these using the original a, b, c. First, notice that

$$a^3 = -\alpha^3 - \beta^3 - \gamma^3 - 3\alpha^2\beta - 3\alpha^2\gamma - 3\alpha\beta^2 - 3\alpha\gamma^2 - 3\beta^2\gamma - 3\beta\gamma^2 - 6\alpha\beta\gamma$$
$$ab = -\alpha^2\beta - \alpha\beta^2 - \alpha^2\gamma - \alpha\gamma^2 - \beta^2\gamma - \beta\gamma^2 - 3\alpha\beta\gamma$$
$$b^3 = \alpha^3\beta^3 + \alpha^3\gamma^3 + \beta^3\gamma^3$$
$$+ 3\alpha^3\beta^2\gamma + 3\alpha^3\beta\gamma^2 + 3\alpha^2\beta^3\gamma + 3\alpha^2\beta\gamma^3 + 3\alpha\beta^3\gamma^2 + 3\alpha\beta^2\gamma^3 + 6\alpha^2\beta^2\gamma^2$$
$$abc = \alpha^3\beta^2\gamma + \alpha^3\beta\gamma^2 + \alpha^2\beta^3\gamma + \alpha^2\beta\gamma^3 + \alpha\beta^3\gamma^2 + \alpha\beta^2\gamma^3 + 3\alpha^2\beta^2\gamma^2$$

Hence,

$$a' = a^3 - 3ab + 3c$$
$$b' = b^3 - 3abc + 3c^2$$
$$c' = c^3$$

Thus our cubic is

$$y = x^3 + (a^3 - 3ab + 3c)x^2 + (b^3 - 3abc + 3c^2)x + c^3$$

A4

A line can only intersect another line at most once; otherwise, they are parallel, which clearly doesn't work if we want our line to intersect all four lines.

Our four points are $(a, 1, 0)$, $(0, b, 1)$, $(1, 0, c)$, $\left(d, d, -\frac{d}{6}\right)$. Since two points specify a line, we need to fix two points such that the line passes through the remaining two. Using the first two points, we have the vector $(-a, b-1, 1)$ and we can choose the first point $(a, 1, 0)$ as origin. Thus, the line's parametric equation is

$$L = (-a, b-1, 1)t + (a, 1, 0)$$

We want to intersect the third line, hence we equate

$$(1, 0, c) = (-a, b-1, 1)t + (a, 1, 0)$$

Thus, $t = c$, from which we derive

$$1 = -ac + a \quad \Rightarrow \quad a = \frac{1}{1-c}$$

$$0 = (b-1)c + 1 \quad \Rightarrow \quad b = 1 - \frac{1}{c}$$

Our line is refined

$$L = \left(-\frac{1}{1-c}, -\frac{1}{c}, 1\right)t + \left(\frac{1}{1-c}, 1, 0\right)$$

To intersect the fourth line, we equate again

$$\left(d, d, -\frac{d}{6}\right) = \left(-\frac{1}{1-c}, -\frac{1}{c}, 1\right) t + \left(\frac{1}{1-c}, 1, 0\right)$$

Thus, $t = -\frac{d}{6}$, from which we derive

$$d = \frac{1 + \frac{d}{6}}{1-c}$$

$$d = 1 + \frac{d}{6c}$$

$$\Rightarrow c = \frac{-1 \pm 5}{12}$$

i.e. $c = -\frac{1}{2}$ or $c = \frac{1}{3}$. The corresponding lines are

$$L = \left(-\frac{2}{3}, 2, 1\right) t + \left(\frac{2}{3}, 1, 0\right) = (-2, 6, 3)\, t + \left(\frac{2}{3}, 1, 0\right)$$

$$L = \left(-\frac{3}{2}, -3, 1\right) t + \left(\frac{3}{2}, 1, 0\right) = (-3, -6, 2)\, t + \left(\frac{3}{2}, 1, 0\right)$$

Chapter 3

Putnam 2000

3.1 A1